BCA Question Bank — 50 Practice MCQs

Covering C Programming, Data Structures, DBMS, Operating Systems & Networking

SECTION 1: PROGRAMMING IN C (Q1 - Q10)

Q1. Who is known as the father of the C Programming Language?

(A) Bjarne Stroustrup
(B) Dennis Ritchie
(C) James Gosling
(D) Ken Thompson

Answer: (B) Dennis Ritchie — Developed C at Bell Labs in 1972.

Q2. What is the size of an `int` data type in standard 32-bit C compilers?

(A) 2 Bytes
(B) 4 Bytes
(C) 8 Bytes
(D) 1 Byte

Answer: (B) 4 Bytes — Standard 32-bit systems allocate 4 bytes (32 bits) for integers.

Q3. Which operator is used to access the memory address of a variable in C?

(A) * (Asterisk)
(B) & (Ampersand)
(C) -> (Arrow)
(D) . (Dot)

Answer: (B) & (Ampersand) — Addressof operator (&) returns the memory address.

Q4. Which header file is required to use `malloc()` and `calloc()` functions?

(A) <stdio.h>
(B) <stdlib.h>
(C) <string.h>
(D) <math.h>

Answer: (B) <stdlib.h> — Standard library header contains dynamic memory allocation routines.

Q5. What will be the output of: `printf("%d", 5 / 2);` ?

(A) 2.5
(B) 2
(C) 3
(D) Compilation Error

Answer: (B) 2 — Integer division truncates the fractional part.

Q6. Which keyword is used to prevent modifications to a variable in C?

(A) static
(B) const
(C) volatile
(D) register

Answer: (B) const — Declares read-only variables.

Q7. What value is returned by default if a function does not return a value explicitly?

(A) 0
(B) 1
(C) Garbage Value
(D) NULL

Answer: (C) Garbage Value — In C, unreturned function values default to garbage.

Q8. What does `calloc()` initialize allocated memory blocks to?

(A) Garbage values
(B) Zero (0)
(C) NULL
(D) One (1)

Answer: (B) Zero (0) — Unlike malloc, calloc initializes all bytes to zero.

Q9. Which string function is used to concatenate two strings in C?

(A) strcpy()
(B) strcat()
(C) strlen()
(D) strcmp()

Answer: (B) strcat() — Appends source string to destination string.

Q10. What is a pointer that points to a memory location that has been deleted or freed called?

(A) NULL Pointer
(B) Dangling Pointer
(C) Void Pointer
(D) Wild Pointer

Answer: (B) Dangling Pointer — Points to invalid/deallocated memory.

SECTION 2: DATA STRUCTURES & ALGORITHMS (Q11 - Q20)

Q11. Which data structure works on the LIFO (Last In First Out) principle?

(A) Queue
(B) Stack
(C) Linked List
(D) Tree

Answer: (B) Stack — Push and Pop occur at the same end (TOP).

Q12. What is the time complexity of Binary Search in the worst case?

(A) O(N)
(B) O(log N)
(C) $O(N^2)$
(D) O(1)

Answer: (B) O(log N) — Halves search space in every step.

Q13. Which data structure is best suited for Breadth-First Search (BFS) graph traversal?

(A) Stack
(B) Queue
(C) Priority Queue
(D) Array

Answer: (B) Queue — FIFO order manages level-by-level processing.

Q14. In a Binary Search Tree (BST), which traversal yields values in ascending sorted order?

(A) Preorder
(B) Inorder
(C) Postorder
(D) Level Order

Answer: (B) Inorder — Left -> Root -> Right yields sorted output.

Q15. What condition occurs when attempting to remove an element from an empty Stack?

(A) Overflow
(B) Underflow
(C) Garbage Value
(D) Segmentation Fault

Answer: (B) Underflow — Pop operation on top == -1 results in Underflow.

Q16. What is the worst-case time complexity of Quick Sort?

(A) O(N log N)
(B) $O(N^2)$
(C) O(N)
(D) O(log N)

Answer: (B) $O(N^2)$ — Occurs when pivot selection leads to highly unbalanced partitions.

Q17. Which node in a Single Linked List has its pointer set to NULL?

(A) First Node
(B) Last Node
(C) Middle Node
(D) Head Pointer

Answer: (B) Last Node — Indicates the end of the linked list.

Q18. What is the minimum number of stacks required to implement a Queue?

(A) 1
(B) 2
(C) 3
(D) 4

Answer: (B) 2 — One for enqueue (push) and one for dequeue (pop).

Q19. Which sorting algorithm is known as stable and guarantees O(N log N) in all cases?

(A) Quick Sort
(B) Merge Sort
(C) Bubble Sort
(D) Selection Sort

Answer: (B) Merge Sort — Divide and conquer guarantees O(N log N).

Q20. What is a full binary tree with height *h* having total nodes formula?

(A) $2^h - 1$
(B) $2^h + 1$
(C) $2^{h+1} - 1$
(D) 2^{h+1}

Answer: (C) $2^{h+1} - 1$ — Maximum nodes in binary tree of height h.

SECTION 3: DATABASE MANAGEMENT SYSTEMS (Q21 - Q30)

Q21. What does SQL stand for?

(A) Structured Query Language
(B) Sequential Query Language
(C) Server Quality Language
(D) Standard Query Logic

Answer: (A) Structured Query Language — Standard domain-specific language for relational databases.

Q22. Which SQL command is classified under DDL (Data Definition Language)?

(A) SELECT
(B) INSERT
(C) CREATE
(D) UPDATE

Answer: (C) CREATE — Defines database schema (CREATE, ALTER, DROP).

Q23. A key that uniquely identifies each record in a database table and cannot be NULL is:

(A) Foreign Key
(B) Primary Key
(C) Candidate Key
(D) Super Key

Answer: (B) Primary Key — Uniquely identifies rows; prevents duplicates and NULL values.

Q24. Which normal form eliminates partial dependencies in database tables?

(A) 1NF
(B) 2NF
(C) 3NF
(D) BCNF

Answer: (B) 2NF — 2NF ensures every non-prime attribute is fully functionally dependent on primary key.

Q25. What does the 'A' in ACID properties of database transactions stand for?

(A) Availability
(B) Atomicity
(C) Authenticity
(D) Accuracy

Answer: (B) Atomicity — All-or-nothing execution principle of database transactions.

Q26. Which SQL clause is used to filter records after aggregate functions are applied?

(A) WHERE
(B) HAVING
(C) GROUP BY
(D) ORDER BY

Answer: (B) HAVING — HAVING filters aggregated groups; WHERE filters rows.

Q27. What command is used to permanently save changes made by a transaction?

(A) ROLLBACK
(B) COMMIT
(C) SAVEPOINT
(D) GRANT

Answer: (B) COMMIT — Saves changes to the database permanently.

Q28. A field in a table pointing to the Primary Key of another table is called a:

(A) Alternate Key
(B) Foreign Key
(C) Composite Key
(D) Unique Key

Answer: (B) Foreign Key — Establishes referential integrity between tables.

Q29. What architecture level in DBMS describes how data is physically stored on disk?

(A) External Level
(B) Conceptual Level
(C) Internal Level
(D) Logical Level

Answer: (C) Internal Level — Physical level dealing with disk blocks and indexing.

Q30. Which join returns all matching records from both tables plus non-matching records?

(A) Inner Join
(B) Full Outer Join
(C) Left Join
(D) Right Join

Answer: (B) Full Outer Join — Combines results of both Left and Right outer joins.

SECTION 4: OPERATING SYSTEMS (Q31 - Q40)

Q31. Which component of the OS remains permanently in main memory (RAM)?

(A) Shell
(B) Kernel
(C) Compiler
(D) File Manager

Answer: (B) Kernel — Core part of OS managing hardware resources.

Q32. Which CPU scheduling algorithm operates strictly on a FIFO basis?

(A) SJF
(B) FCFS
(C) Round Robin
(D) Priority Scheduling

Answer: (B) FCFS — First-Come, First-Served processes in arrival order.

Q33. What is the time quantum associated with?

(A) FCFS Scheduling
(B) Round Robin Scheduling
(C) Priority Scheduling
(D) Multilevel Queue

Answer: (B) Round Robin Scheduling — Fixed time slice allocated per process.

Q34. What is a deadlock prevention algorithm designed by Edsger Dijkstra?

(A) Round Robin
(B) Banker's Algorithm
(C) Page Replacement
(D) Peterson's Algorithm

Answer: (B) Banker's Algorithm — Avoids deadlock by testing resource allocations for safety.

Q35. What is the memory management scheme that allows process physical address space to be non-contiguous?

(A) Segmentation
(B) Paging
(C) Dynamic Linking
(D) Swapping

Answer: (B) Paging — Maps fixed-size pages to physical frame blocks.

Q36. What occurs when a requested page is not currently present in RAM?

(A) Page Fault
(B) Segmentation Error
(C) Deadlock
(D) Cache Miss

Answer: (A) Page Fault — OS must fetch required page from secondary storage.

Q37. Belady's Anomaly occurs in which page replacement algorithm?

(A) LRU
(B) FIFO
(C) Optimal
(D) LFU

Answer: (B) FIFO — Increasing frames can paradoxically increase page faults in FIFO.

Q38. What is a light-weight unit of CPU execution within a process called?

(A) Micro-process
(B) Thread
(C) Semaphore
(D) Task

Answer: (B) Thread — Shares process memory space while running independently.

Q39. What integer variable is used to solve critical section synchronization problems?

(A) Mutex
(B) Semaphore
(C) Monitor
(D) Spinlock

Answer: (B) Semaphore — Non-negative integer used for process synchronization.

Q40. What state is a process in when it is waiting for an I/O operation to complete?

(A) Ready
(B) Running
(C) Blocked / Waiting
(D) Terminated

Answer: (C) Blocked / Waiting — Process cannot proceed until I/O finishes.

SECTION 5: NETWORKING & COMPUTER BASICS (Q41 - Q50)

Q41. How many layers are present in the OSI Reference Model?

(A) 4
(B) 5
(C) 7
(D) 6

Answer: (C) 7 — Physical, Data Link, Network, Transport, Session, Presentation, Application.

Q42. Which layer of the OSI model is responsible for routing IP packets?

(A) Data Link Layer
(B) Network Layer
(C) Transport Layer
(D) Physical Layer

Answer: (B) Network Layer — Handles logical IP addressing and packet routing.

Q43. What is the standard size of an IPv4 address?

(A) 32 Bits (4 Bytes)
(B) 64 Bits
(C) 128 Bits
(D) 16 Bits

Answer: (A) 32 Bits (4 Bytes) — e.g., 192.168.1.1 is 32 bits long.

Q44. Which protocol translates Domain Names (e.g., google.com) into IP addresses?

(A) DHCP
(B) DNS
(C) FTP
(D) HTTP

Answer: (B) DNS — Domain Name System maps human names to IP addresses.

Q45. What default port number is used by secure HTTP (HTTPS)?

(A) 80
(B) 443
(C) 21
(D) 22

Answer: (B) 443 — HTTP uses port 80; HTTPS uses port 443.

Q46. Which topology connects all network nodes to a single central hub or switch?

(A) Bus Topology
(B) Ring Topology
(C) Star Topology
(D) Mesh Topology

Answer: (C) Star Topology — Central device manages data distribution to all nodes.

Q47. What protocol provides connection-oriented, reliable transport service?

(A) UDP
(B) TCP
(C) IP
(D) ICMP

Answer: (B) TCP — Transmission Control Protocol ensures error-checked ordered delivery.

Q48. Which device operates at Layer 2 (Data Link Layer) using MAC addresses?

(A) Router
(B) Switch
(C) Hub
(D) Gateway

Answer: (B) Switch — Switches forward frames based on MAC addresses.

Q49. What protocol automatically assigns dynamic IP addresses to devices on a network?

(A) DNS
(B) DHCP
(C) ARP
(D) SNMP

> **Answer: (B) DHCP** — Dynamic Host Configuration Protocol automates IP allocation.

Q50. What is the standard size of an IPv6 address?

(A) 32 Bits
(B) 64 Bits
(C) 128 Bits
(D) 256 Bits

> **Answer: (C) 128 Bits** — IPv6 uses 128-bit hexadecimal addresses.